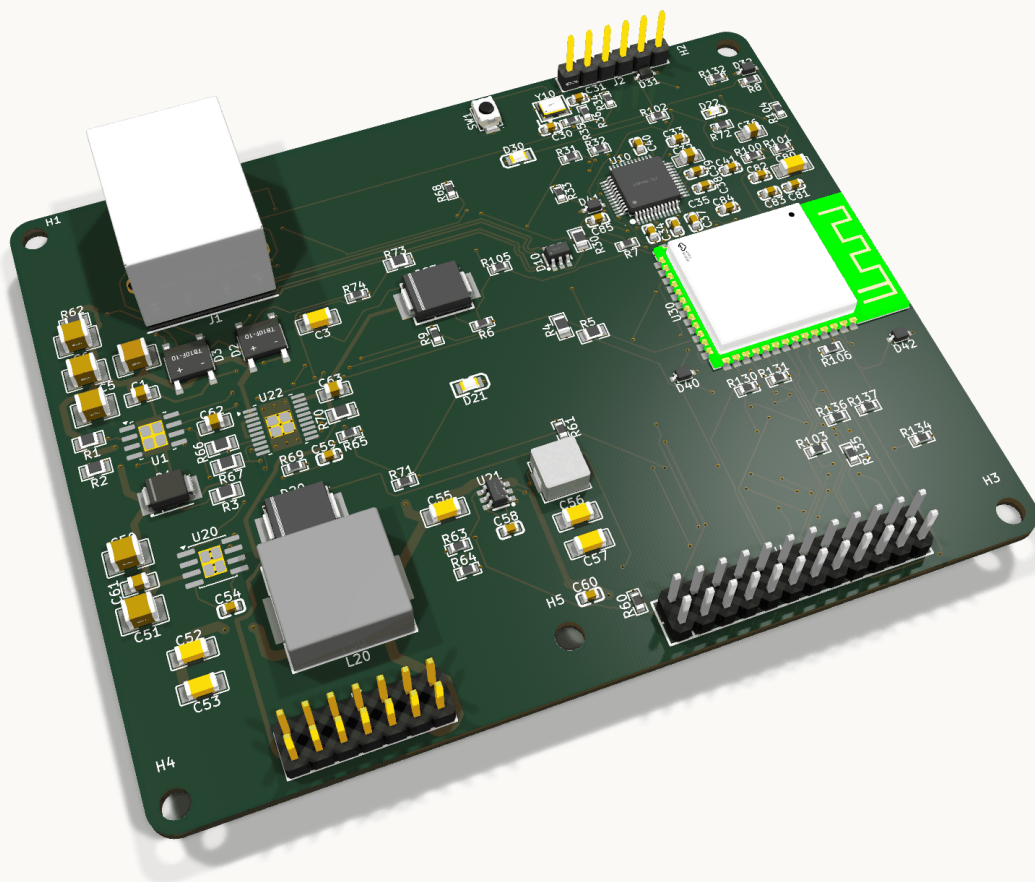


hwde

Printed circuit boards, designed from a written brief.

You describe the board in plain words. hwde researches the parts, picks a topology, draws the schematic, lays out and routes the board, runs the electrical and manufacturing checks, and hands you the files a factory needs. A human engineer signs off at every stage.



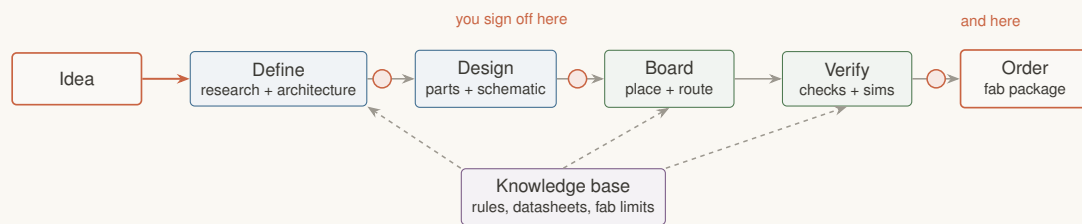
LUMINA-CARRIER 4 LAYERS · 100 × 80 MM · 116 FOOTPRINTS ORDERED, FABRICATED

A carrier board for a stage-lighting fixture that hwde designed.

How it works

Three pictures follow. The first is the route a board takes, the second is who does which part of the work, and the third is what happens at every handover.

FIGURE 1 *the route a board takes*

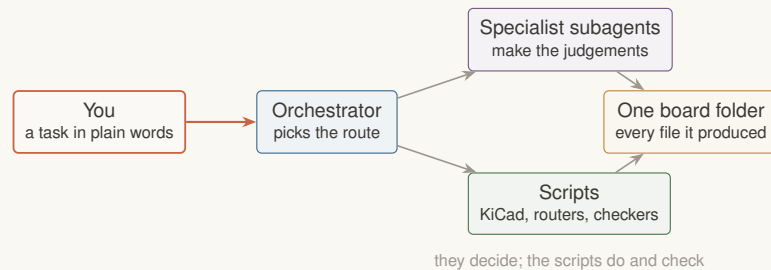


Each box in the row is a stage, and the box below it is the knowledge base that the dashed arrows read. The orange dots mark where a person has to look at the work and agree before it goes on.

A run starts from a written brief and ends with a manufacturing package of Gerbers, a bill of materials and a pick-and-place file. Each stage in between produces one file, and before it does, it reads the knowledge base of design rules, datasheet numbers and what the factory can really make.

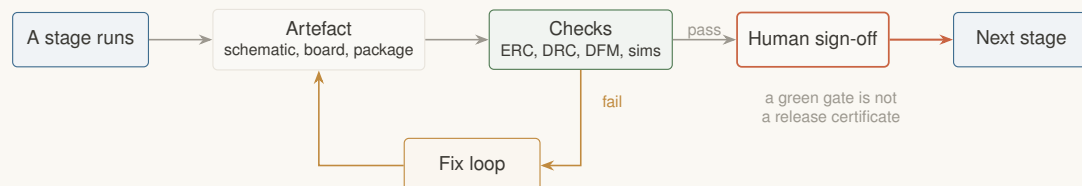
Who does what

The work is split in two. Twenty-two specialist subagents make the judgement calls, like choosing a topology, sizing a part or reviewing a schematic, and each one has a written contract for what it may decide. About sixty scripts do and check the mechanical work, and they are all deterministic. They drive KiCad, run the routers and export the Gerbers. A script never has an opinion, and an agent never edits a board file by hand. Whatever either of them produces lands as a file in one folder per board.

FIGURE 2 *who does what*

The orange box is you and the blue one routes the work. Purple makes the judgement calls, green does the mechanical work, and both write to the board folder.

The gate loop

FIGURE 3 *what happens at every handover*

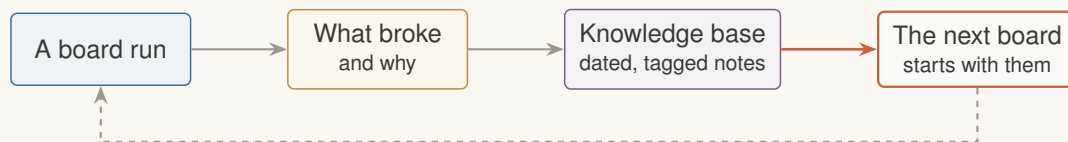
The green box is the checks and the orange one is a person's sign-off. The line from the checks down to the fix loop is the fail path, and it returns to the file the stage made.

Every stage ends at a gate. The gate runs the checks that apply to that stage, which are the electrical rules, the design rules, manufacturability and circuit simulations against stated bounds. It either passes, or it hands the findings back to a fix loop that reworks the file and tries again. A pass is not the end of it, because a person still signs off before the next stage starts.

Figures 2 and 3: docs/showcase/diagrams/roles.tex and gate.tex. The checks are the scripts in `.claude/skills/hwde/scripts`.

How it gets better

FIGURE 4 *how one run feeds the next*



Solid arrows are one board run, from what broke to the notes it leaves. The dashed line back is the next board starting from those notes.

Every run writes down what broke and why, with a date and a tag, and that goes into the knowledge base the next run reads first. The rule is to record a gotcha the moment you hit it, so the next board does not pay for it again.

hwde takes a board from a written brief to the files a factory needs, and a person signs off at every stage.

WHAT HWDE IS FOR

WHAT A PASS DOES NOT MEAN

A green gate only proves the checks that actually ran, so the reports are kept one per check instead of being collapsed into a verdict. The stage a board reached is not a release certificate either. A person decides what is fit to release.

Figure 4: docs/showcase/diagrams/lessons.tex. The notes are LEARNINGS.md.

The boards

Fourteen boards got far enough to have a laid-out, routed PCB. They run from a linear regulator with five parts to a 200 W RF power stage.

TABLE 1 *the fourteen boards and where each one stopped*

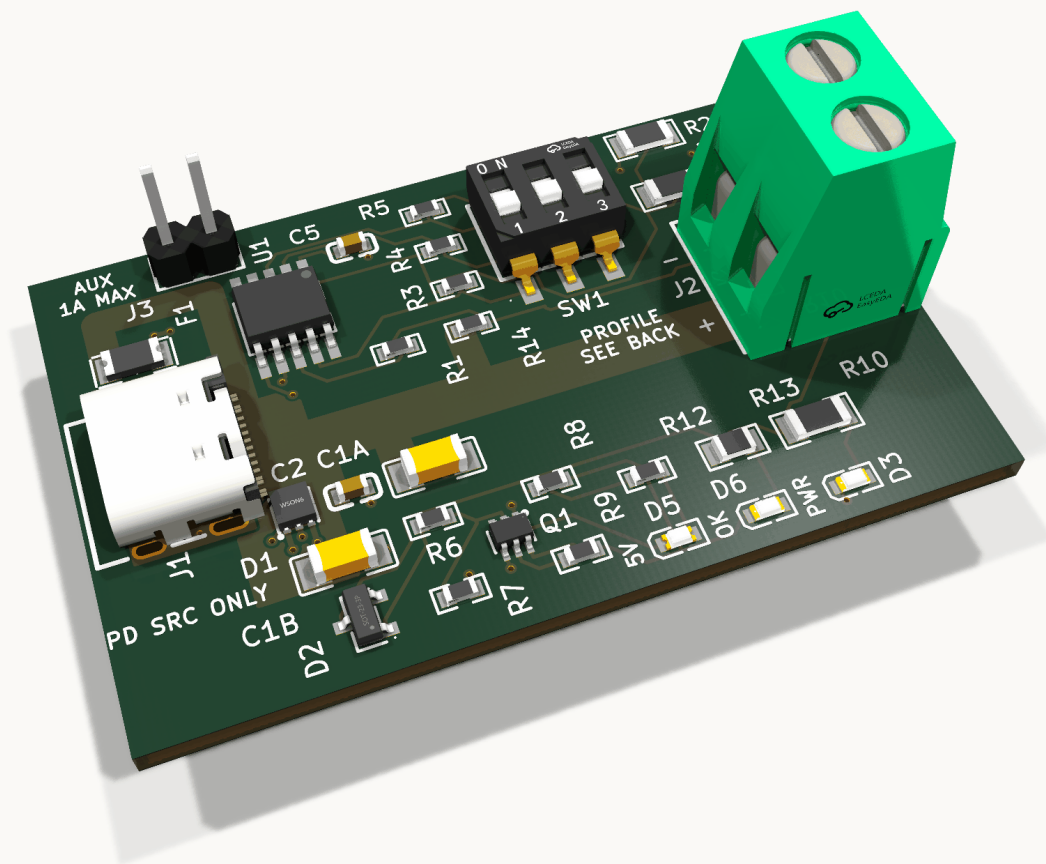
BOARD	WHAT IT IS	LAYERS	SIZE (MM)	STAGE REACHED
lumina-carrier	lighting carrier, PoE-powered	4	100 × 80	ordered, fabricated
pd-trigger	USB-C PD bench trigger	2	48 × 30	ordered, fabricated
lumina-par	RGBW PAR daughter board	4	100 × 80	verification
rf-de-20m	20 MHz Class E GaN stage	4	120 × 80	manufacturability
sbuck-5v3a	5 V / 3 A synchronous buck	4	50 × 40	package ready
usb-buck	USB device dev board	4	50 × 40	package ready
g0-sense	USB-C sensor node	2	35.8 × 28.3	package ready
stm32-blinky	minimal STM32 dev board	2	50 × 40	package ready
rf-term-150w	150 W RF dummy load head	2	26 × 20	manufacturability
bb-adc	single-channel ADC	2	54.8 × 34.9	manufacturability
bb-amp	sensor amplifier chain	2	48 × 28.3	manufacturability
bb-buck	bare buck converter	2	35 × 25	manufacturability
bb-ldo	bare linear regulator	2	34.7 × 34.7	manufacturability
bb-mcu	bare microcontroller board	2	34.8 × 22.3	manufacturability

“Stage reached” says where the board stopped in the pipeline, and it does not say the board is fit to release. Two boards say “ordered, fabricated” because their own files record a JLCPCB order number that tracked through to shipped. The rest stopped with a checked manufacturing package in hand. Each size is the bounding box of the board outline, and not every outline is a rectangle.

Read 23 September 2026 from boards/<board>/kicad/<board>.kicad_pcb (layers, size) and boards/<board>/state.json (stage).

Designed, ordered and fabricated

Every render below was made from the board file in the repository, by the script that sits next to this document. `lumina-carrier`, on the first page, is the universal carrier for a stage-lighting fixture. It takes power over Ethernet through a PD controller and a 100 V buck, and it carries a frozen expansion connector that a fixture-specific daughter board plugs into. It is the biggest board here, with 4 layers and 116 footprints, and another board depends on the connector it sets. The other board that was ordered is a bench tool.

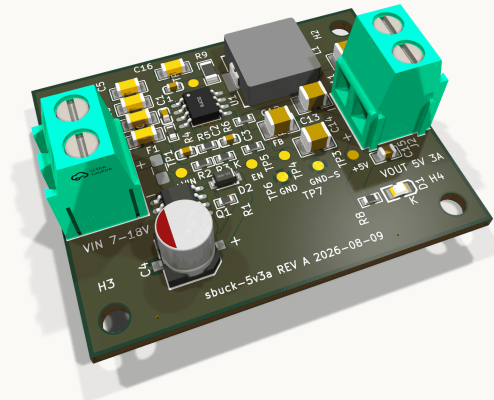


PD-TRIGGER 2 LAYERS · 48 × 30 MM · 30 FOOTPRINTS **ORDERED FROM JLCPCB, TRACKED TO SHIPPED**

A USB-C Power Delivery sink controller negotiates with a charger for 5, 9, 12, 15 or 20 volts, and you pick the profile on the board. The negotiated rail comes out on a screw terminal rated for 5 A, so up to 100 W. The board converts nothing. It just passes the rail through.

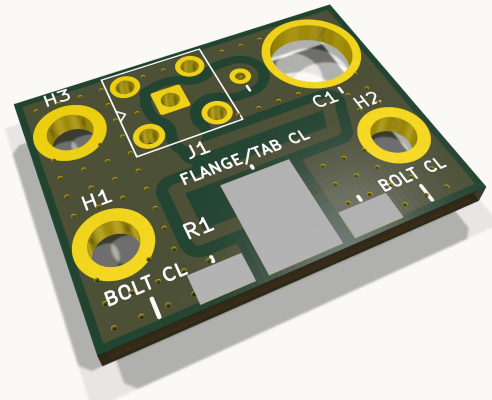
Rendered by `docs/showcase/render_boards.sh` (KiCad 10.0.5) from the board file in `boards/<board>/kicad`, for `pd-trigger`. Layers, size and footprints from the same file, the stage from `state.json`, 23 September 2026.

Power



SBUCK-5V3A 4 LAYERS · 50 × 40 MM · 44
FOOTPRINTS **PACKAGE READY**

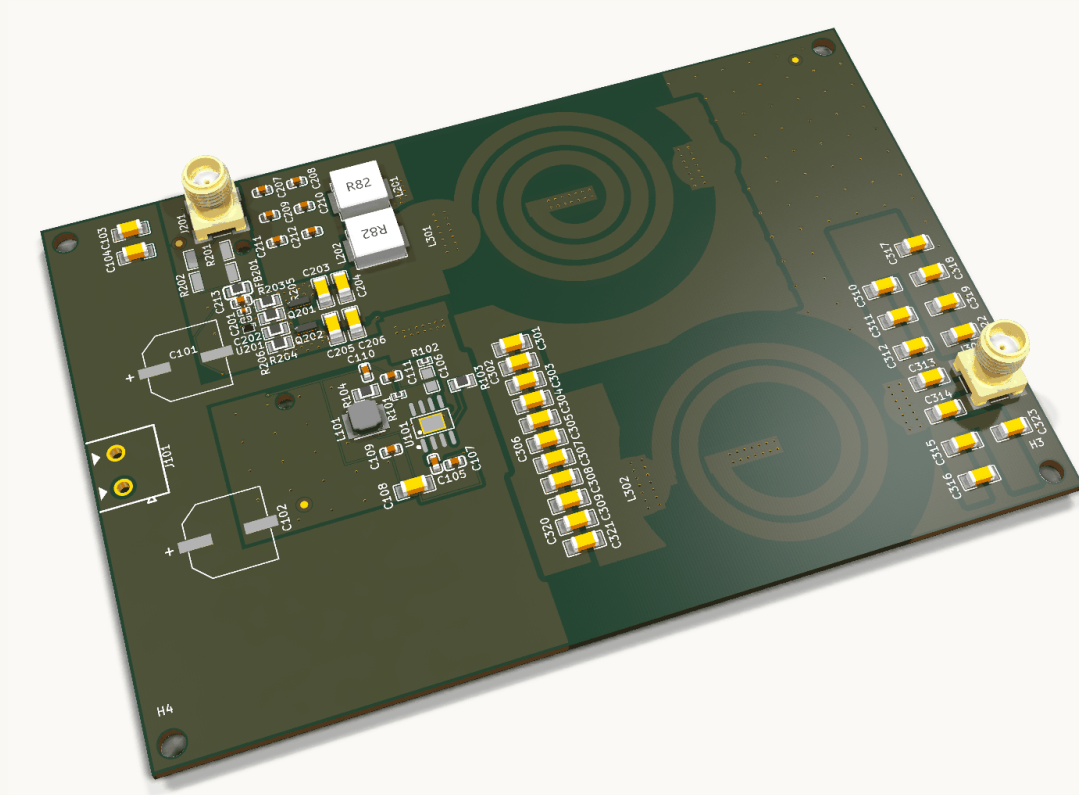
A synchronous buck that takes 7–18 V in and holds 5.0 V $\pm$ 2% at 3 A out. It is an open-frame power module.



RF-TERM-150W 2 LAYERS · 26 × 20 MM · 6
FOOTPRINTS **MANUFACTURABILITY**

A 50-ohm, 150 W dummy load head for DC to 25 MHz. The element bolts to a heatsink the user supplies, so the board is only the RF launch and the mechanical interface. It renders close to bare because none of its six footprints has a 3D model the renderer could find, so the connector and the element are not drawn.

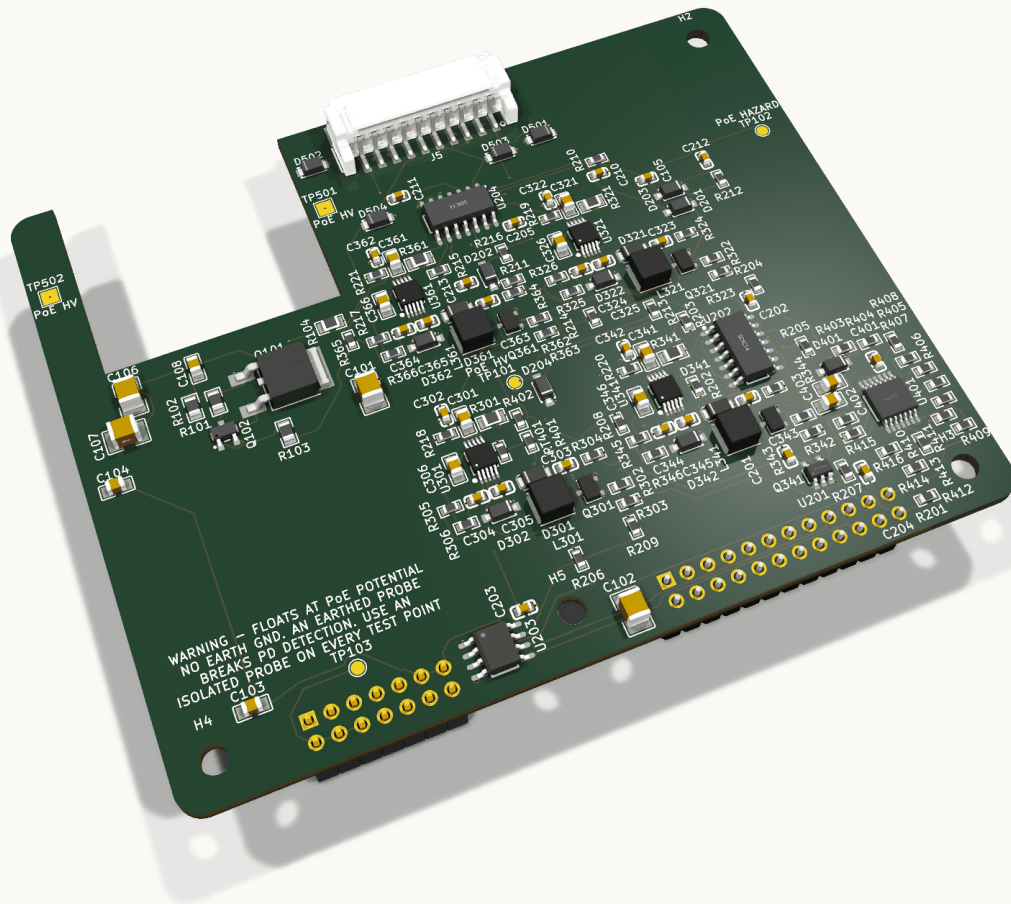
Harder boards



RF-DE-20M 4 LAYERS · 120 × 80 MM · 80 FOOTPRINTS STOPPED AT MANUFACTURABILITY CHECKS

A 20 MHz Class E RF power stage that puts 200 W into 50 ohms from a 40 V bus. A PWM drive arrives on an SMA input, a GaN gate driver buffers it, and one ground-referenced eGaN FET switches. A resonant tank and an L-match transform the 4.6-ohm load line to 50 ohms at the output SMA. The owner fixed the topology and the two core parts after a costed trade study. hwde did the rest.

Rendered by docs/showcase/render_boards.sh (KiCad 10.0.5) from the board file in boards/<board>/kicad, for rf-de-20m. Layers, size and footprints from the same file, the stage from state.json, 23 September 2026.

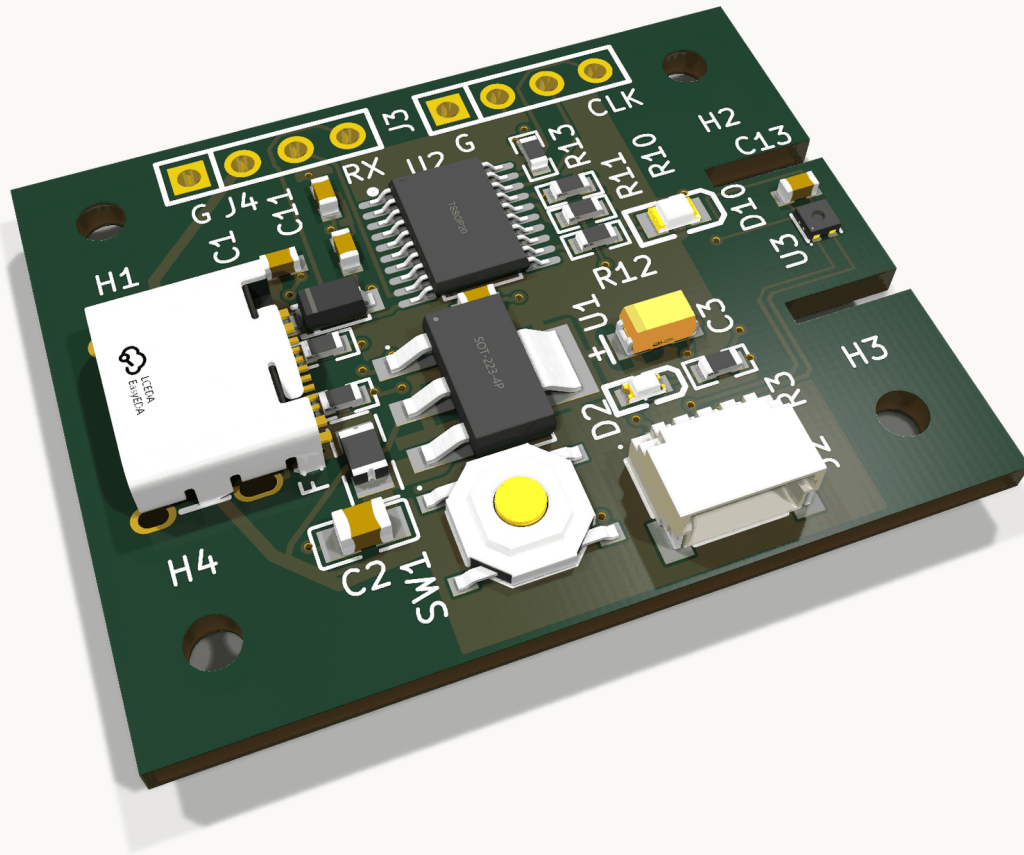


LUMINA-PAR 4 LAYERS · 100 × 80 MM · 158 FOOTPRINTS STOPPED AT VERIFICATION

The RGBW PAR daughter board that stacks on the carrier. The carrier's connector document is frozen, so this board takes it as given and may not redefine it. Its power budget of 8.6 to 20 W was re-derived here, and depends on the power-over-Ethernet class the fixture gets. It is the densest board here at 158 footprints, and the only one whose outline is not a rectangle. The silkscreen warns that the board floats at the power-over-Ethernet potential, because an earthed probe breaks the fixture's power negotiation.

Rendered by docs/showcase/render_boards.sh (KiCad 10.0.5) from the board file in boards/<board>/kicad, for lumina-par. Layers, size and footprints from the same file, the stage from state.json, 23 September 2026.

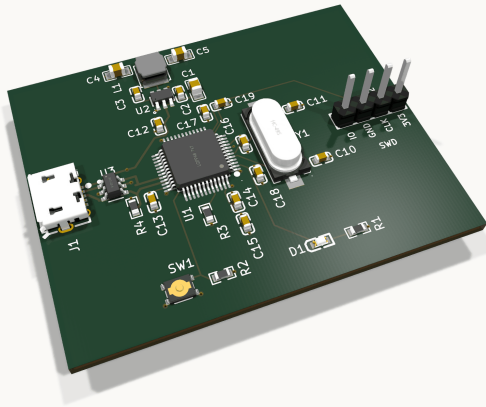
Microcontroller boards



G0-SENSE 2 LAYERS · 35.8 × 28.3 MM · 30 FOOTPRINTS **PACKAGE READY, RELEASE ATTESTATION BUILT**

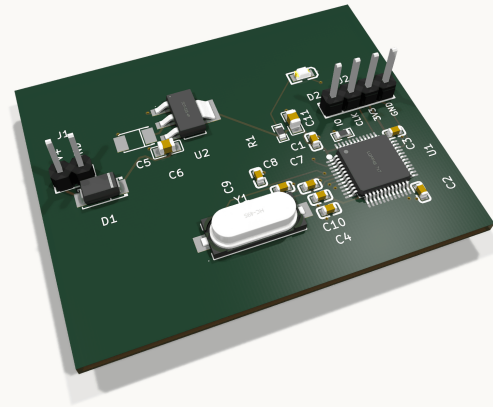
A temperature and humidity node that runs off USB-C. USB-C supplies 5 V and an LDO makes 3.3 V from it, and an STM32G030 on its internal oscillator reads a Sensirion SHT4x over I²C. The same bus comes out on a Qwiic connector, the readings go out over a UART header, and you program it over SWD. hwde designed this one end to end in a single unattended container run.

Rendered by docs/showcase/render_boards.sh (KiCad 10.0.5) from the board file in boards/<board>/kicad, for g0-sense. Layers, size and footprints from the same file, the stage from state.json, 23 September 2026.



USB-BUCK 4 LAYERS · 50 × 40 MM · 28
FOOTPRINTS PACKAGE READY

An STM32F103 board that talks USB at full speed. The Micro-B socket carries both power and data, and there is an LED, a button and an SWD header.

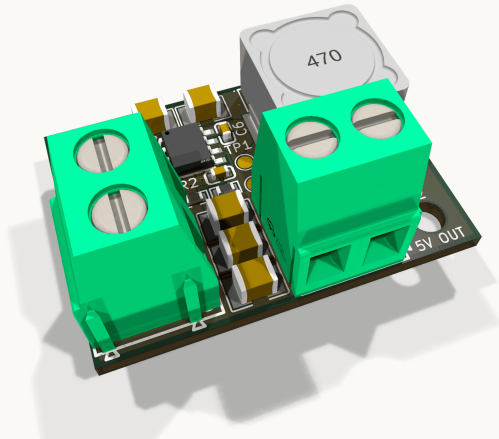


STM32-BLINKY 2 LAYERS · 50 × 40 MM · 20
FOOTPRINTS PACKAGE READY

The simplest useful board here. It has an STM32F103, an 8 MHz crystal, one LED, an LDO and an SWD header.

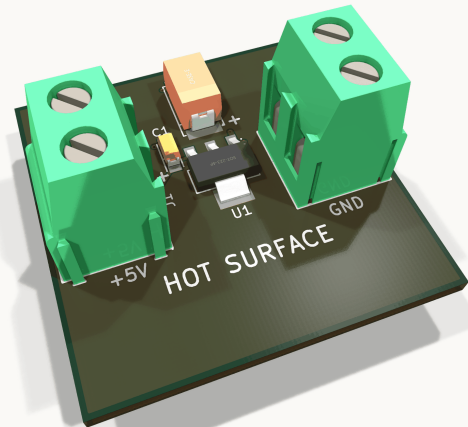
Building blocks

These five boards are deliberately bare, each one carrying a single block and nothing else. They are there to be studied and measured on a bench, so they leave out protection, filtering, indicators and spare rails on purpose. All five stopped at manufacturability checks.



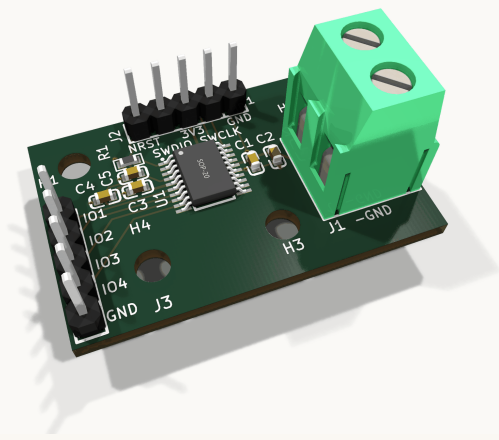
BB-BUCK 2 LAYERS · 35 × 25 MM · 20
FOOTPRINTS

It takes 18–30 V in and gives 5 V at 2 A out on a screw terminal. It carries the converter and exactly what its datasheet requires.



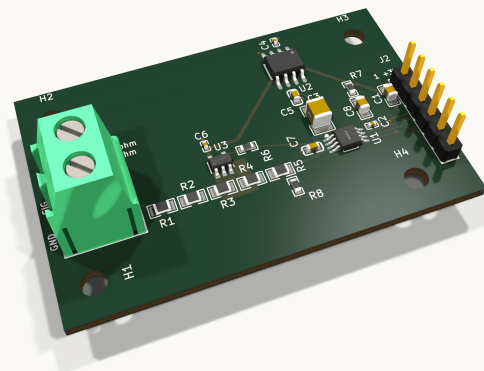
BB-LDO 2 LAYERS · 34.7 × 34.7 MM · 5
FOOTPRINTS

It takes 5 V in and gives 3.3 V at 500 mA out, holding regulation in still air on the board's own copper.



BB-MCU 2 LAYERS · 34.8 × 22.3 MM · 14
FOOTPRINTS

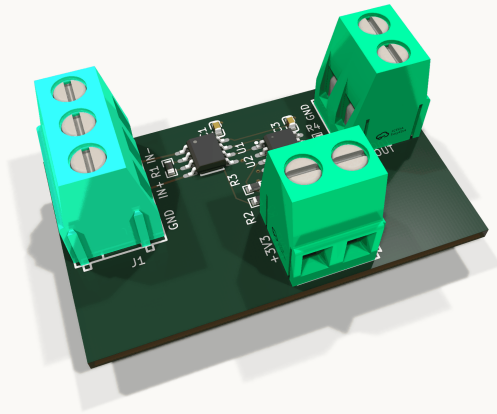
One MCU with a 3.3 V rail coming in, its debug header and four GPIO. There is no regulator and no second rail.



BB-ADC 2 LAYERS · 54.8 × 34.9 MM · 25
FOOTPRINTS

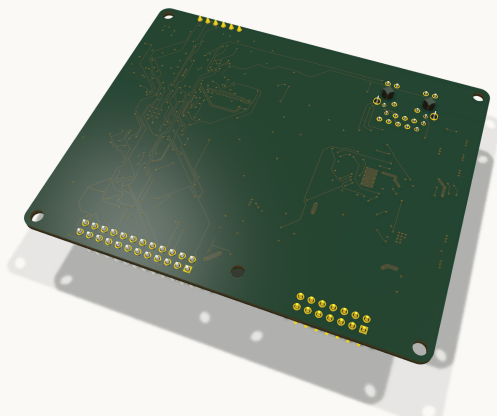
0–5 V in on a screw terminal, and 12 bits or better at up to 10 kSa/s out over a 0.1-inch header. It trades speed for DC accuracy.

Rendered by docs/showcase/render_boards.sh (KiCad 10.0.5) from the board file in boards/<board>/kicad, for bb-buck, bb-lldo, bb-mcu and bb-adc. Layers, size and footprints from the same file, the stage from state.json, 23 September 2026.



BB-AMP 2 LAYERS · 48 × 28.3 MM · 14
FOOTPRINTS

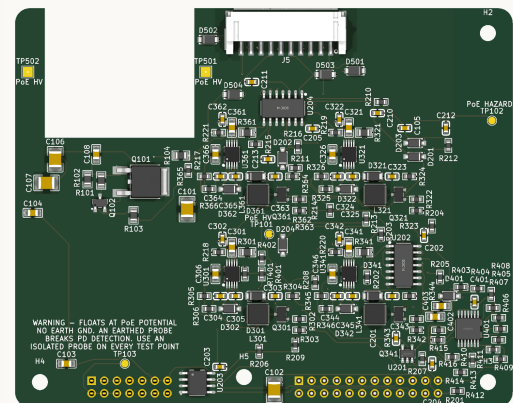
It takes a 0–20 mV differential bridge signal and gives 0–3.3 V single-ended out, from DC to 1 kHz. The nominal gain is 165 V/V.



LUMINA-CARRIER, UNDERNEATH BOTTOM
VIEW

Every part sits on the top side of this board, so the back carries routing and vias and nothing else. This is how the factory sees it.

Two more boards stopped before layout and have no PCB to show, a 5 V/3 A buck and a strobe daughter board. Both are in the repository at the stage they reached.



LUMINA-PAR, STRAIGHT DOWN TOP-DOWN
VIEW

The daughter board on page 9, from straight above, so the routing shows. It has four layers and 158 footprints, routed to the same rules as everything else.

Rendered by docs/showcase/render_boards.sh (KiCad 10.0.5) from the board file in boards/<board>/kicad, for bb-amp, lumina-carrier and lumina-par. Layers, size and footprints from the same file, the stage from state.json, 23 September 2026.

What it can and cannot do yet

hwde is an engineering assistant, and it does not release a board by itself. A human engineer drives it and signs off at every stage.

14	2	200 W	158
boards with a routed PCB	ordered from JLCPCB and fabricated	out of the RF stage, into 50 ohms	footprints on lumina-par, the densest

What holds

- Fourteen boards went from a written brief to a routed layout, including a 4-layer 200 W RF stage and a novel 100 W USB-C PD design.
- Two of them were designed with it, ordered from a factory and fabricated, and their own files record the order numbers tracked through to shipped.
- Every check is a script, and they all follow the same contract. The findings are kept per check rather than as a single verdict.
- Everything a run produces is a file in that board's folder, and you can read it without hwde.

What does not

- The factory has no public API for its manufacturability review, so someone still does that step by hand in a browser.
- Spending money is deliberately hard. One code path can place an order, and it handles 2-layer boards only. It refuses a board whose files already record an order, and it will not run without a fresh quote and a matching design hash. A person also has to type out a confirmation naming the board, the quantity and the total.
- The electrical and thermal checks are engineering screens, and each states its own accuracy. None of them is a certification, and nothing here is safety-certified for mains, medical, automotive or aerospace use.

The numbers are Table 1 and the board files behind it. The order path is the ordering subagent and its scripts in `.claude/skills/hwde`.